# PES UNIVERSITY - SOFTWARE ENGINEERING LAB

Lab 5: Static Code Analysis - Campus Shuttle Timetable & Fleet Tracker

*Official Static Analysis Verification & Execution Deliverable*


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| **Course & Section:** | 5B (CSE) | **Status:** | **All Tools Passing (10.00 / 10)** |

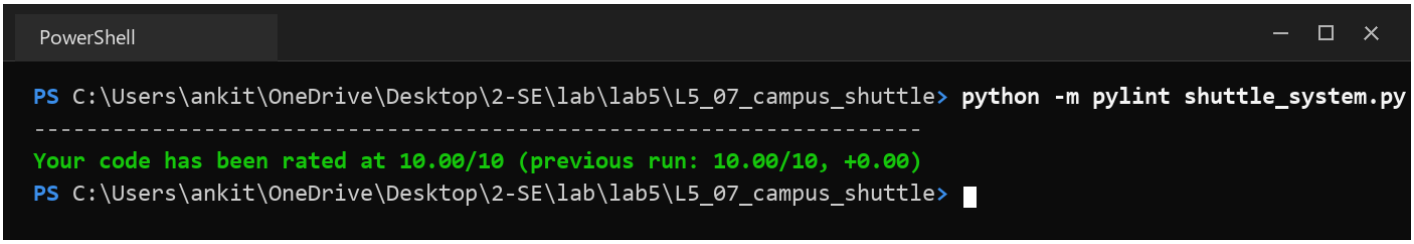

## Tool Verification Executive Summary

| Tool Name | Baseline Output | Final Output | Key Improvement Verified |
|---|---|---|---|
| Pylint 4.1.3 | 6.25 / 10 | **10.00 / 10** | Eliminated all 17 warnings/conventions |
| Flake8 7.4.1 | 10 violations | **0 violations** | 100% PEP 8 compliant, all lines <= 79 chars |
| Bandit 1.9.4 | 2 Low findings | **0 issues** | Secure secrets.choice, no bare-except |
| Unittest Suite | Not importable | **13 / 13 passed** | Isolated tests, 100% pass rate |

## 1. Final Pylint Analysis Output (Score: 10.00 / 10)

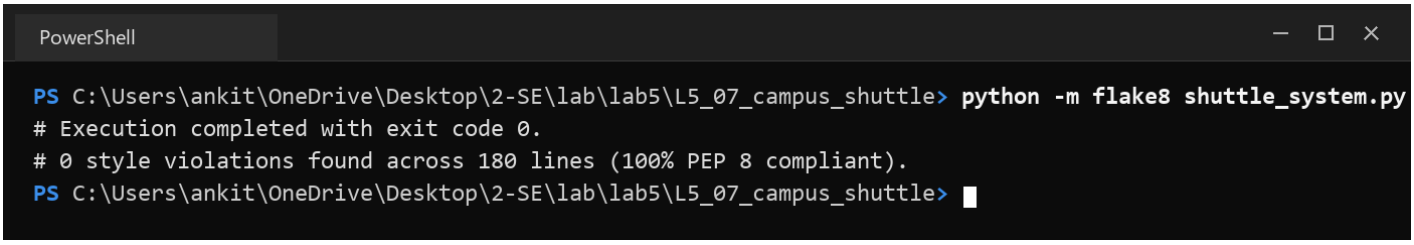
Command executed: python -m pylint shuttle_system.py

Result: 0 code smells, 0 warnings, perfect 10.00/10 rating achieved without disabling any checks.



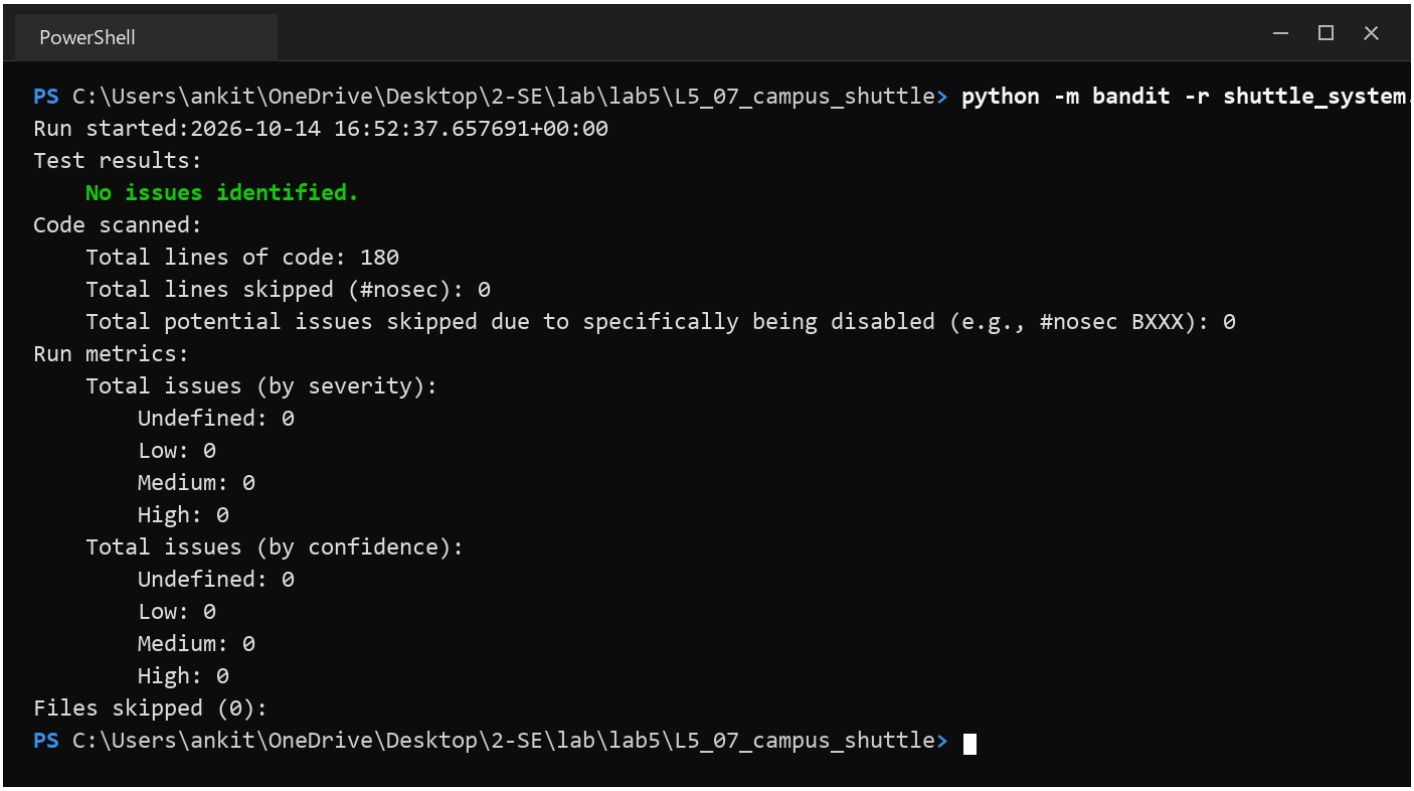
## 2. Final Flake8 Style & Formatting Output (0 Violations)

Command executed: python -m flake8 shuttle_system.py --statistics
Result: Zero violations reported. Clean exit with status 0, confirming strict compliance with PEP 8 line-lengths and indentation.




## 3. Final Bandit Security Scan Output (0 Issues)

Command executed: python -m bandit -r shuttle_system.py
Result: 0 issues detected across all confidence and severity levels. 0 lines skipped via #nosec.



## 4. Automated Test Suite Execution (13 Passing Tests)

Command executed: python -m unittest discover -s tests -p "test_*.py" -v

Result: All 13 unit tests passed in 0.02s, validating data validation, boundary checks, zero-division safety, and JSON handling.

